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INTERIM FEDERAL SPECIFICATION

W-H-00150(GSA-FSS)

HEATER, COOLANT, ENGINE (ELECTRIC)

This Interim Federal Specification was developed by Standardization Division, Federal Supply Service, General Services Administration, Washington 25, D. C., based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers an electrically actuated device for heating the coolant in a gasoline or diesel engine cooling system when the engine is not operating. Essentially, they are heaters consisting of a housing containing an immersion heating element arranged to be permanently attached to the engine and connected to the cooling system, or they may consist of heaters to be applied to the outside of or inserted in the engine block; as specified and required to facilitate quick and easy starting of the engine in below freezing temperature.

1.1.1 Federal specification coverage. - Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover only the types, classes, grades, and sizes which are generally purchased by the Federal Government.

1.2 Classification.

1.2.1 Grades, types, classes, and sizes. - The engine cooling system preheater shall be of the following grades, types, classes, and sizes as specified in the invitation to bid, contract or order (see 6.1).

1.2.2 Grades. - The preheater shall be of one grade for operation on alternating-current.

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1.2.3 Types. - The preheater shall be either type I or type II as specified (see 6.1):

Type I. - With thermostatic temperature control.
Type II. - With or without temperature control.

1.2.4 Classes and sizes. - The preheater shall be of the following classes and sizes as specified (see 6.1):

Class A. - Heaters installed in block or hose, without positive circulator.
Size 300. - 300-cubic inch engine displacement, 400 watts minimum.

Class B. - Heaters installed exterior to block or hose, with positive circulator.

Size	300.	- 300-	" " " "	750	watts minimum.
Size	600.	- 600-	" " " "	1000	" "
Size	800.	- 800-	" " " "	1500	" "
Size	1200.	- 1200-	" " " "	2250	" "
Size	1600.	- 1600-	" " " "	3000	" "
Size	2000.	- 2000-	" " " "	4000	" "

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 Specifications and standards. - There are no other Federal specifications applicable to the specification. The following standards, of the issue in effect on date of invitation to bid, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Standards:
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Nongovernmental:
National Board of Fire Underwriters Pamphlet:
No. 499 - Standards for Electrical Heating Appliances.

(Copies may be obtained from the National Board of Fire Underwriters, 85 John Street, New York 38, N. Y.; 222 West Adams St., Chicago 6, Ill.; or 465 California St., San Francisco 4, Calif.)

Underwriters' Laboratories, Inc., Standards:

(Copies may be obtained from Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 1655 Scott Lane, Santa Clara, Calif.).

Consolidated Classification Committee Publication:
Consolidated Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Underwriters' Laboratories Inc., Standards.

3.1.1 The bidder shall submit proof to the Purchasing Agency that the appliance or components thereof, he proposes to supply under this specification conform to the standards of the Underwriters' Laboratories, Inc., regarding fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as conforming to this requirement.

3.1.2 In lieu of the label or listing, the bidder may submit independent proof satisfactory to the Purchasing Agency that his appliance or the component parts thereof, conform to the published standard, including method of test of the Underwriters' Laboratories, Inc.

3.1.3 Compliance with the above preliminary requirements, regarding fire and casualty hazards, does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure the acceptance of his material or appliance.

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3.2 Design and construction. - The preheater shall consist of an electrical heating element capable of heating tap water, used as an engine coolant, to the test temperatures herein specified for the applicable engine capacity. Unless otherwise specified, (see 6.1), the preheater shall be for use with 115 or 230 volt electricity and shall be of the minimum capacities herein specified for the respective sizes.

3.2.1.1 Type I. - Each type I preheater shall be provided with thermostatic control of the electric heater switches and to maintain engine cooling water temperatures under the conditions hereinafter specified for respective capacities of classes A and B. The thermostat may be an integral part of the preheater, or a separate unit for mounting in other locations of the water system.

3.2.1.2 Type II. - For type II preheaters thermostatic control of the engine cooling water is not required.

3.2.2 Classes. - The preheater shall be either of the immersion type designed for permanent attachment to the engine, or a separate unit for installation outside of the block within the water coolant circulation system, as specified for classes A and B (see 1.2.4 and 6.1).

3.2.2.1 Class A preheaters. - The class A preheaters shall consist of a heater designed for insertion in the engine block or hose. For class A, the heater functions by contact heating of the coolant of the water as it surrounds the heater or passes it by gravity flow. Positive means of circulation of the heated water is not required to be furnished with the heater.

3.2.2.2 Class B preheaters. - The class B preheater shall be separately housed and designed for operation outside of the engine block or hose. It shall consist of a housing(s) containing the heating element and equipment or fittings to insure positive means for circulating the engine coolant.

3.3 Capacity.

3.3.1 Class A. - The class A preheater shall have a heating element of sufficient capacity to maintain a temperature of not less than 50°F. with an ambient temperature of 0°F. in an engine cooling system having a total piston displacement of that specified for the size heater indicated in the invitation to bid, contract or order.

3.3.1.1 Class A, size 300. - The class A, size 300 heater shall preheat the cooling system of an engine with 300-cubic inch total piston displacement by method of a heating element of not less than 400 watts minimum capacity when operating on 115 or 230 volt current.

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3.3.2 Class B. - The class B preheater shall have a heating element of sufficient capacity to maintain a temperature of not less than 50°F. with an ambient temperature of minus 10°F. in an engine cooling system having a total piston displacement of that specified for the size heater indicated in the invitation to bid, contract, or order (see 6.1).

3.3.2.1 Class B, size 300. - The class B, size 300 heater shall preheat the cooling system of an engine with 300-cubic inch total piston displacement by means of a heating element of 750 watts minimum capacity, when operating on 115 or 230 volt current.

3.3.2.2 Class B, size 600. - The class B, size 600 heater shall preheat the cooling system of an engine with 600-cubic inch total piston diplacement by means of a heating element of 1000 watts minimum capacity when operating on 115 or 230 volt current.

3.3.2.3 Class B, size 800. - The class B, size 800 heater shall preheat the cooling system of an engine with 800-cubic inch total piston displacement by means of a heating element of 1500 watts minimum capacity, when operating on 115 or 230 volt current.

3.3.2.4 Class B, size 1200. - The class B, size 1200 heater shall preheat the cooling system of an engine with 1200-cubic inch total piston displacement by means of a heating element of 2250 watts minimum capacity, when operating on 115 or 230 volt current.

3.3.2.5 Class B, size 1600. - The class B, size 1600 heater shall preheat the cooling system of an engine with 1600-cubic inch total piston displacement by means of a heating element of 3000 watts minimum capacity when operating on 115 or 230 volt current.

3.3.2.6 Class B, size 2000. - The class B, size 2000 heater shall preheat the cooling system of an engine with 2000-cubic inch total piston displacement by means of a heating element of 4000 watts minimum capacity when operating on 115 or 230 volt current.

3.4 Material. - The material shall be adequate in its physical properties for the purpose intended; and shall be resistant to corrosion by its composition or surface plating or treatment.

3.4.1 Fastening devices. - All screws, pins, bolts, and similar parts shall be installed with adequate means for preventing loss of proper tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.2 Electrical connections. - Electrical connections shall be either mechanically secured, soldered, or brazed. Use of soldering acids, acid fluxes, or soldering salts shall be avoided whenever practicable. If used, all joints or terminals shall be thoroughly cleaned and all traces of acid removed.

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3.4.3 Safety devices. - The preheater shall be furnished with suitable safety devices of latest type. Ample protection against electrical shock shall be provided.

3.4.4 Interchangeability. - All replaceable parts shall be constructed to definite standards, tolerances, and clearances in order that any such part of a particular type or model may be replaced or adjusted without requiring modification. All such parts, where practicable, shall be permanently and legibly marked with the manufacturer's part number.

3.5 Electrical equipment.

3.5.1 Voltage and current. - The heaters shall be designed for operation on the voltage and current characteristics specified (see 6.1).

3.5.2 Extension cord. - Each heating unit shall have an approved type of flexible rubber covered appliance cord, with wires equivalent to not less than 16 gage, and not less than 3 feet long.

3.6 Painting. - Unless otherwise specified, the painting and finish for the heater shall be the manufacturer's standard (see 6.1).

3.7 Instruction for operation and care. - A handbook of instructions for operation and care shall be furnished with each heater supplied under this specification.

3.8 Spare parts. - Spare parts, as specified, shall be furnished with each heater (see 6.1).

3.9 Workmanship. - The heater shall be new and of the quality workmanship necessary to produce a heater of good appearance, having stability and efficient operating and productive characteristics as found in first-class commercial-quality items of a similar nature.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General. - The heater and all the material entering into the manufacture thereof, shall be subject to inspection by authorized Government inspector. When inspection is conducted at the contractor's plant, all tests required by this specification and/or contract or purchase order shall be conducted by the contractor under the supervision of an authorized Government inspector. The Government inspector may at his option conduct any or all the tests. The place of inspection and acceptance shall be as specified in the invitation for bids, contract, or purchase order (see 6.1).

4.2 Sampling. - Samples for inspection and tests shall be taken from each lot or shipment on the basis of one sample from a lot of 10 or less, two samples from 11 to 50, and for lots of over 50, two samples, plus one for each additional hundred or fraction thereof.

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4.2.1 Inspection. - Each heater from an inspection lot shall be given a general inspection for material, workmanship, dimensions, performance, weight, painting, marking, packaging, packing, labeling, and general compliance with the requirements of this specification.

4.3 Performance tests.

4.3.1 The heater of each size shall be tested for heating capacity by connecting it to an engine block or a water tank or container of equivalent cubic inch engine displacement as required by this specification and observing its ability to maintain the required temperature, under the specified conditions for one hour.

5. PREPARATION FOR DELIVERY
(Nonmilitary agencies. - Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 Level A. - Level A preservation and packaging shall afford adequate protection against corrosion, deterioration, and physical damage during shipment, handling, indeterminate storage and world-wide redistribution.

5.1.2 Level C. - Level C preservation and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 Level A. - Level A packing shall afford adequate protection during shipment, handling indeterminate storage, and world-wide redistribution.

5.2.2 Level B. - Level B packing shall afford adequate protection against multiple domestic shipments, handling and storage.

5.2.3 Level C. - Commodities preserved and packaged as specified in 5.1.1 and 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules, or regulations of other carriers, or regulations of oceangoing vessels as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military requirements. - On military procurement, interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

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5.3.2 Nonmilitary requirements. - In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size heater required (see 1.2, 3.2 to 3.3.2 inclusive.)
(c) Type I with, or type II without thermostatic control (see 3.2.1, 3.2.1.1 and 3.2.1.2).
(d) Make and model of vehicle with which the heater will be used.
(e) Voltage and current requirements (see 3.5.1).
(f) Color of paint and type of finish, if different (see 3.6).
(g) Spare parts, if required (see 3.8).
(h) Place of inspection and acceptance (see 4.1).
(i) Level of preservation, packaging and packing required (see section 5).
(j) Marking, if different (see 5.3).

6.1.1 Data with bid. - Each bidder should be required to furnish with his bid, complete descriptive data covering the items, he proposes to furnish under this specification and the invitation for bid.

6.2 Guarantees. - The invitation for bids should require the following performance and life guarantees to become a part of the contract.

6.2.1 Performance guarantee. - The heater will operate satisfactorily in accordance with the requirements of this specification on the vehicle identified in the invitation to bid, contract or order.

6.2.2 Life guarantee. - The heater is guaranteed for the period indicated below:

Class A heaters are guaranteed for a period of 6 months from the date of delivery, against defective material or workmanship, and further to replace at point of installation, if within the continental United States, or f.o.b., the continental United States port to be designated by the purchasing officer if installation is outside the continental United States, promptly without cost to the Government, all parts proving defective within the guarantee period.

Class B heaters are guaranteed for a period of 12 months from the date of delivery, against defective material or workmanship, and further to replace at point of installation, if within the continental United States or f.o.b. the continental United States port to be designated by the purchasing officer if installation is outside the continental United States, promptly without cost to the Government, all parts proving defective within guarantee period.

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Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation, whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not be be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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